Câu 1

[2] Create a database, then write SQL statements to create, within this database, all tables derived from the ERD shown in the following picture, with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as requested, you MUST keep the table names, relationships, attributes and data types of attributes the SAME as given in the ERD.

- Attributes with underline belong to the Primary Key of each entity.
- Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.
- When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do **not** use "create database", "After database", "use database_name" or any statements using database's name in your submission.

Entities and attributes shown in **Picture 1.1**:

- **contract** (contractID varchar(70), quantity int)
- **Contracts** — M relationship between contract and another entity
- **Cost-of**, **Rents-of**, **Service-of** — relationships
- **Customers** (custID varchar(50), custAddress nvarchar(200), custName nvarchar(100))
- **Rooms** (roomID varchar(50), roomPrice money)
- **Services** (serviceID varchar(50), serviceName nvarchar(100), servicePrice money)

Câu 2

[1] Write a query to get the `model` and `year` of the car(s) whose `colour` is `'Black'`, as shown in **Picture 2.1**:

#	model	year
1	volkswagen	2017
2	ford	2017
3	buick	2017
4	Grand Caravan	2017
5	dodge	2017

Câu 3

[1] Write a query to display `custName`, `phone`, `sex`, `cusAddress`, and `NumberOfInvoiceID` for each customer. Only display customers whose `NumberOfInvoiceID` **is greater than or equal to 3**.

Example result (**Picture 3.1**):

#	custName	phone	sex	cusAddress	NumberOfInvoiceID
1	Đoàn Minh Đạo	879355621	M	Huế	3
2	Phan Thành Đăng	12272718	M	Khánh Hòa	3

Câu 4

[1] Write a query to display `custName`, `model`, `serviceTicketID`, `dateReturned` of the car(s) whose `dateReturned` falls in the first quarter of 2021.

Example result (**Picture 4.1**):

#	custName	model	serviceTicketID	dateReturned
1	Phạm Nguyên Dũng	dodge	10111243	2021-01-07
2	Đoàn Minh Đạo	audi	10111239	2021-03-16
3	Hồ Đức Trung Hữu	chevrolet	10111237	2021-03-20
4	Nguyễn Thị Minh Hà	toyota	10111235	2021-03-29
5	Phạm Nguyên Dũng	chevrolet	10111231	2021-02-07

Câu 5

[1] Write a query to display the mechanic with the most working hours in 2021. The result includes columns: `mechanicName`, `sumHours`.

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Example result (**Picture 5.1**):

#	mechanicName	sumHours
1	Nguyễn Văn Linh	8

Câu 6

[1] Create a stored procedure named `proc_serviceTicket_part` that counts the distinct parts belonging to a given service ticket, with `@serviceTicketID int` as an input parameter and `@numberOfParts int` as an output parameter.

Example: when the procedure `proc_serviceTicket_part` is executed using the following statements, the result should be as in **Picture 6.1**:

```
declare @x int
exec proc_serviceTicket_part 10111236, @x output
select @x
```

Result: `3`

Câu 7

[1] Create a trigger named `tr_insert_car` for the insert statement on table `Cars` so that when one car is inserted into `Cars`, the system displays all cars whose `year` and `model` are the same as those of the inserted car.

The result includes: `carID`, `serialNumber`, `model`, `colour`, `year`.

Example: when the following statement is executed, the system will display the results as in **Picture 7.1**:

```
insert into Cars(carID, serialNumber, model, colour, year)
values (11223344101, '83ABC83ABC83ABC83', 'toyota', 'Black', 2017)
```

Expected result:

#	carID	serialNumber	model	colour	year
1	1122334475	13ABC13ABC13ABC13	toyota	Gray	2017
2	11223344101	83ABC83ABC83ABC83	toyota	Black	2017

(Note: the preview image is only a sample; the result depends on entered data.)

Câu 8

[0.5] Create a stored procedure named `proc_salesPerson_invoice` that counts the distinct invoices belonging to a given sales person, with `@salesID decimal` as an input parameter and `@numberOfInvoices int` as an output parameter.

Example: when the procedure `proc_salesPerson_invoice` is executed using the following statements, the result should be as in **Picture 8.1**:

```
declare @t int
declare @a decimal(18,0)
set @a = 30121050035
exec proc_salesPerson_invoice @a, @t output
select @t
```

Result: `4`

Câu 9

[1] Write a query to delete all rows with `serviceID = 12` in the table `ServiceMehanic`.

Ví dụ (**Picture 9.1**) — các dòng sẽ bị xóa (trích):

#	serviceTicketID	serviceID	mechanicID	hours	comment	rate
26	10111240	2	41051230031	2	NULL	2000000.00
27	10111240	7	41051230244	4	NULL	1650000.00
28	10111240	16	41051230041	1	NULL	1000000.00
29	10111241	3	41051230032	2	NULL	1800000.00

30	10111241	5	41051230026	2	NULL	2000000.00
31	10111241	11	41051230060	1	NULL	7600000.00
32	10111242	16	41051230158	5	NULL	3200000.00
33	10111243	13	41051230034	6	NULL	800000.00
34	10111244	3	41051230025	1	NULL	1800000.00
35	10111244	10	41051230142	2	NULL	6500000.00
36	10111245	12	41051230038	2	NULL	1000000.00

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Câu 10

[0.5] Create a function named `f_tk`. This function takes an input parameter `(@mechanicID decimal(18,0))` and returns a table. The returned table includes `mechanicID`, `mechanicName`, `sumHours`. Note: `sumHours` is the total working hours of the given mechanic's id.

Example: when the function `f_tk` is executed using the following statement, the result should be as in **Picture 10.1**:

```
select * from f_tk(41051230025)
```

Expected result:

#	mechanicID	mechanicName	sumHours
1	41051230025	Cao Xuân Trường	6